

1. Apriori Algorithm and FP-Growth

Given: Support = 3, Confidence = 50%.

Software : WEKA

Steps

1. Open **WEKA**.
2. Select **Explorer**.
3. Go to **Preprocess**.
4. Open the required .arff dataset.
5. Click **Associate**.
6. Select **Apriori**.
7. Set minimum confidence to **50%**.
8. Set the required minimum support.
9. Click **Start**.
10. Record the generated frequent itemsets/rules.
11. For FP-Growth, select **FPGrowth** instead of Apriori.
12. Set the same required support/confidence.
13. Click **Start**.
14. Record the generated rules.

Program code : .arff file

```
@relation market_basket

@attribute M {0,1}
@attribute O {0,1}
@attribute N {0,1}
@attribute K {0,1}
@attribute E {0,1}
@attribute Y {0,1}
@attribute D {0,1}
@attribute A {0,1}
@attribute U {0,1}
@attribute C {0,1}
@attribute I {0,1}

@data
1,1,1,1,1,1,0,0,0,0,0
0,1,1,1,1,1,1,0,0,0,0
1,0,0,1,1,0,0,1,0,0,0
1,0,0,1,0,1,0,0,1,1,0
0,1,0,1,1,0,0,0,0,1,1
```

OUTPUT :

The screenshot displays the Weka GUI Chooser interface. The 'Program' tab is selected, and the 'Apriori' algorithm is chosen from the 'Associate' category. The 'Apriori output' window is open, showing the results of the association rule mining process.

Apriori output

```
=== Apriori model (full training set) ===  
Apriori  
=====
```

Minimum support: 0.85 (4 instances)
Minimum metric <confidence>: 0.5
Number of cycles performed: 3

Generated sets of large itemsets:

Size of set of large itemsets L(1): 6
Size of set of large itemsets L(2): 6
Size of set of large itemsets L(3): 1

Best rules found:

1. B=1 & ==> B=1 & <conf: (1)> lift: (1) lev: (0) [0] conv: (0)
2. D=0 & ==> B=1 & <conf: (1)> lift: (1) lev: (0) [0] conv: (0)
3. B=0 & ==> B=1 & <conf: (1)> lift: (1) lev: (0) [0] conv: (0)
4. D=0 & ==> B=1 & <conf: (1)> lift: (1) lev: (0) [0] conv: (0)
5. L=0 & ==> B=1 & <conf: (1)> lift: (1) lev: (0) [0] conv: (0)
6. D=0 & ==> B=1 & <conf: (1)> lift: (1.25) lev: (0.14) [0] conv: (0.8)
7. B=1 & ==> D=0 & <conf: (1)> lift: (1.25) lev: (0.14) [0] conv: (0.8)
8. B=1 D=0 & ==> B=1 & <conf: (1)> lift: (1) lev: (0) [0] conv: (0)
9. B=1 D=0 & ==> B=1 & <conf: (1)> lift: (1.25) lev: (0.14) [0] conv: (0.8)
10. B=1 B=1 & ==> D=0 & <conf: (1)> lift: (1.25) lev: (0.14) [0] conv: (0.8)

Status
OK

Log

Applications

- Explorer
- Experimenter
- KnowledgeFlow
- Workbench
- Simple CLI

Weka Environment for Knowledge Discovery
Version 3.8.7
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